

# Slide 1: Title

## Resume Based AI Chatbot using Gemini API

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Technology Stack:

- Python
  - Gemini API
  - Prompt Engineering
  - LLM (Large Language Model)
- 

## Slide 2: Project Objective

### Objective

Develop a personal AI chatbot that can:

- Answer questions about my resume
  - Explain skills, projects, education, and experience
  - Generate intelligent responses using Gemini AI
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## Slide 3: Complete Code Structure

```
from google import genai

client = genai.Client(api_key="YOUR_API_KEY")

my_data = """ Resume Information """

print("Mukund AI Chatbot Started...")
print("Type 'exit' to quit.")

while True:

    question = input("You: ")

    if question.lower() == "exit":
        print("Bot: Goodbye!")
```

```
        break

    prompt = f"""
    Resume Data
    User Question
    """

    response = client.models.generate_content(
        model="gemini-2.5-flash",
        contents=prompt
    )

    print(response.text)
```

---

## Slide 4: Import Statement

### Step 1: Import Gemini Library

Code:

```
from google import genai
```

Purpose:

- Import Gemini SDK
- Allow Python to communicate with Gemini AI

Without this import, Gemini API cannot be used.

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## Slide 5: API Connection

### Step 2: Create Gemini Client

Code:

```
client = genai.Client(
    api_key="YOUR_API_KEY"
)
```

Purpose:

- Connect application with Gemini server
- Authenticate using API key

Flow:

Python Program ↓ API Key ↓ Gemini Server

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## Slide 6: Resume Data Storage

### Step 3: Store Resume Information

Code:

```
my_data = """
Name: Mukund Kumar
Education
Skills
Projects
Experience
"""
```

Purpose:

Store all personal information inside a variable.

Contains:

- Education
  - Skills
  - Experience
  - Projects
  - Certifications
- 

## Slide 7: Start Messages

### Step 4: Display Startup Message

Code:

```
print("Mukund AI Chatbot Started...")  
print("Type 'exit' to quit.")
```

Output:

Mukund AI Chatbot Started...

Type 'exit' to quit.

Purpose:

Inform user that chatbot is running.

---

## Slide 8: Infinite Loop

### Step 5: Keep Chatbot Running

Code:

```
while True:
```

Purpose:

- Run chatbot continuously
- Accept unlimited questions

Flow:

Question ↓ Answer ↓ Question ↓ Answer

---

## Slide 9: User Input

### Step 6: Take User Question

Code:

```
question = input("You: ")
```

Example:

You: What is your education?

Stored in variable:

question

Purpose:

Receive question from user.

---

## Slide 10: Exit Condition

### Step 7: Stop Chatbot

Code:

```
if question.lower() == "exit":
```

Purpose:

Convert input to lowercase and check:

Examples:

EXIT Exit exit

All become:

exit

---

## Slide 11: Break Statement

### Step 8: Exit Program

Code:

```
print("Bot: Goodbye!")
break
```

Output:

```
Bot: Goodbye!
```

Purpose:

Terminate chatbot loop.

---

## Slide 12: Prompt Engineering

### Step 9: Create Prompt

Code:

```
prompt = f"""
You are Mukund Kumar's personal AI assistant.

Use ONLY the information below to answer questions.

Information:
{my_data}

Question:
{question}
"""
```

Purpose:

Combine:

- Resume Data
- User Question

Into one prompt.

This is called:

Prompt Engineering

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## Slide 13: Actual Prompt Sent to Gemini

Example:

Question:

What is Mukund's education?

Prompt Sent:

You are Mukund Kumar's personal AI assistant.

Information:

B.E CSE  
CGPA 7.8

Question:  
What is Mukund's education?

Purpose:

Provide context to Gemini.

---

## Slide 14: Generate Response

### Step 10: Gemini API Call

Code:

```
response = client.models.generate_content(  
    model="gemini-2.5-flash",  
    contents=prompt  
)
```

Purpose:

Send prompt to Gemini.

Gemini:

- Reads prompt
  - Understands question
  - Generates response
- 

## Slide 15: Response Object

Gemini returns:

```
response
```

Answer stored in:

```
response.text
```

Example:

```
Mukund completed B.E in Computer Science and Engineering from Chandigarh  
University with a CGPA of 7.8.
```

---

## Slide 16: Print Answer

### Step 11: Display Response

Code:

```
print("\nBot:", response.text)
```

Output:

```
Bot: Mukund completed B.E in Computer Science and Engineering from Chandigarh  
University with a CGPA of 7.8.
```

Purpose:



Show answer to user.

---

## Slide 17: Complete Working Flow

User ↓ Input Question ↓ question Variable ↓ Resume Data + Question ↓ Prompt Creation ↓ Gemini API  
↓ Generate Response ↓ response.text ↓ Print Answer

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## Slide 18: Important Concepts Used

1. Python
  2. Gemini API
  3. Prompt Engineering
  4. Context Injection
  5. Large Language Model (LLM)
  6. API Integration
  7. Conversational AI
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## Slide 19: What I Learned

- Gemini API Integration
  - Prompt Engineering
  - AI Chatbot Development
  - LLM Fundamentals
  - Context-Based Responses
  - Python Programming
  - API Communication
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## Slide 20: Future Enhancements

- Resume PDF Upload
- LangChain Integration
- RAG Implementation

- Vector Database
  - Chat Memory
  - Web Interface using Django
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## Slide 21: Demo

Questions:

What is your education?

What projects have you worked on?

What are your skills?

Live Demonstration

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## Slide 22: Thank You

Questions?